Scrapy cram sheet

scrapy/scrapy@2.19.0 · 2026-11-02

If you only read one thing: use `-O out.json` (overwrite), not `-o`. In a project, `-o` onto an existing file logs an ERROR and the items are lost.

Legend: ★ most used · ⚠ gotcha

GET STARTED

```
pip install scrapy
scrapy startproject quotes
cd quotes
scrapy genspider qs site.com
scrapy crawl qs -O out.json
# no project needed:
scrapy runspider s.py \
  -O o.json
```

Needs Python 3.11+.

SPIDER SKELETON

```
import scrapy

class QS(scrapy.Spider):
  name = "qs"
  allowed_domains = [
    "site.com"]
  start_urls = [
    "https://site.com/"]

  def parse(self, r):
    for q in r.css("p"):
      yield {"t": q.css(
        "b::text").get()}
    yield from r.follow_all(
      css="a.next")
```

`follow_all` defaults to `callback=self.parse`. Custom first requests: `async def start(self)` (2.13+).

DAILY COMMANDS

Command	Does
`crawl NAME` ★	run project spider
`shell URL` ★	try selectors live
`list`	spider names
`fetch URL`	print body as Scrapy sees it
`view URL`	open that body in a browser
`parse URL -c parse`	run one callback
`genrequest "curl …"`	curl → `Request(...)`

`crawl`, `list`, `check`, `parse`, `edit` need a project dir.

CRAWL FLAGS

Flag	Effect
`-O f.json` ★	overwrite feed
`-o f.jsonl`	append feed
`-a cat=books`	spider attr `self.cat`
`-s NAME=VAL`	override a setting
`-L INFO`	log level (default DEBUG)
`--pdb`	debugger on failure

SELECTORS

```
r = response
r.css("a::text").get()
  # first match or None
r.xpath("//a/@href").getall()
  # list of all hrefs
r.xpath("//h1/text()").get()
r.follow(href, cb)
  # relative URLs OK
```

DEFAULTS: LIBRARY → NEW PROJECT

- `ROBOTSTXT_OBEY` : False → True
- `CONCURRENT_REQUESTS_PER_DOMAIN` : 8 → 1
- `DOWNLOAD_DELAY` : 0 → 1 s (±50% jitter)
- `FEED_MODE` : None → `create`

A spider run outside a project gets the library column. Check: `scrapy settings --get NAME`

GOTCHAS

- ⚠ Links outside `allowed_domains` drop silently (DEBUG log only), because `OffsiteMiddleware` filters them. Check stat `offsite/filtered`.
- ⚠ A URL seen once is never fetched again: the dupefilter drops it. Use `Request(url, dont_filter=True)`.
- ⚠ Every crawl opens Telnet (:6023) and a RemoteControl HTTP server that runs code. Off: set `TELNETCONSOLE_ENABLED` and `REMOTE_CONTROL_ENABLED` to False.
- ⚠ No-reactor mode (`TWISTED_REACTOR_ENABLED` off) is experimental, uses aiohttp, and can't be set per spider.

TUNE AND RESUME

```
-s JOBDIR=crawls/r1
  # Ctrl-C, rerun to resume
-s HTTPCACHE_ENABLED=1
  # cache: .scrapy/httpcache
-s AUTOTHROTTLE_ENABLED=1
  # delay follows latency
-s CLOSESPIDER_ITEMCOUNT=50
  # stop after 50 items
```

HACK ON SCRAPY

```
pip install -e .
pip install tox pre-commit
tox -e py -- tests/test_x.py
pre-commit run --all-files
tox -e mypy && tox -e pylint
```

All defaults: `default_settings.py` in `scrapy/settings/`. Engine: `scrapy/core/engine.py`.